Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18BS5CS01]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

PROBABILITY, STATISTICS AND RANDOM PROCESSES

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed? **04**

b. In a bolt factory there are four machines A, B, C and D manufacturing respectively 20%, 15%., 25% and 40% of the total production. Out of these 5%, 4%, 3% and 2% are defective. If a bold drawn at random was found defective what the probability that it was manufactured by A is or D? **05**

c. Let X be a continuous random variable with PDF **05**

$$f(x) = \begin{cases} 6x(1-x) & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

and let $Y = X^3$ Find $f_Y(y)$.

2. a. An Airline knows that 5% of the people making reservations on a certain flight will not turn up. Consequently, their policy is to sell 52 tickets for a flight that can hold 50 passengers. What is the probability that there will be a seat for every passenger who turns up? **05**

b. In a partially destroyed laboratory record of an analysis of correlation data the following results are legible $\sigma^2_x = 9$, Regression lines: 8X – 10Y + 66 = 0, 40X – 18Y = 214. Find i) Mean values of X and Y ii) Coefficient of Correlation r **04**

c. Fit a curve of the type $y = ab^x$ to the data **05**

x:	2	3	4	5	6
y:	144	172.8	207.4	248.8	298.6

3. a. A population consist of 4 numbers 3,7, 11, 15 find **04**

i. Mean and S.D of sampling distribution of means by considering sampling of size 2 with replacement

ii. verify that $\mu_{\bar{x}} = \mu$ and $\sigma^2_{\bar{x}} = \frac{\sigma^2}{n} \times \frac{N-n}{N-1}$

b. An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed, with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours. **05**

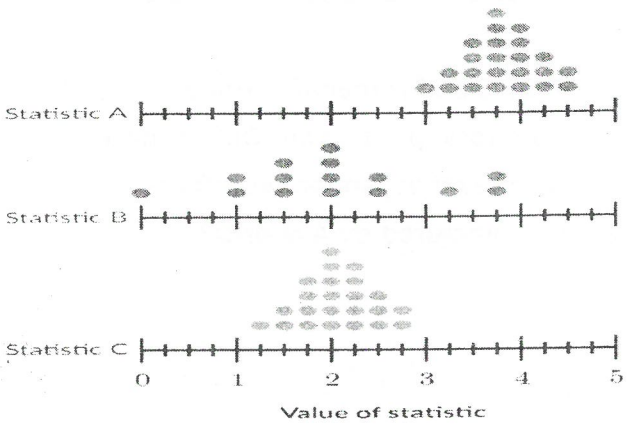
c. Let X1, X2, . . . Xn be a random sample of size n from a distribution with probability density function $f(x,\beta) - \frac{1}{2\beta^3} x^2 e^{-x/\beta}, x > 0, \beta > 0$. Obtain the maximum likelihood estimator of β **05**

OR

4. a. The weight of 1500 ball bearings are normally distributed with mean of 635 gm and S.D of 1.36 gm. If 300 sample of size 36 are drawn from population, find the mean and S.D of sampling distribution of means if sampling is done With replacement. Also find how many samples have their mean greater than 635.5 gm with replacement. **04**

b. Explain the difference between Biased and Unbiasedness Estimator. **05**

The dot plots below show an approximation to the sampling distribution for three different estimators of the same population parameter



If the actual value of the population parameter is 2, which dot plot displays the estimator with low bias and high variability?

c. Let $f(x: \mu, \sigma)$ be given by a Gaussian distribution where x is from our set of n measurements (x_1, x_2,... xn). What is the Maximum likelihood estimation of parameters μ *and* σ. **05**

5. a. An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with mean of 800 hours and S.D of 40 hours. Test hypothesis H0: μ=800 hours against H1: μ≠800. If a sample of 30 bulbs has an average life of 788 hours. Use 5% level of significance. **04**

b. In a sample of 8 observations, the sum of squared deviation from mean is 94.5. in another sample of 10 observation the sum of squared deviation from mean is 101.7. At 5 % level of significance test whether is there any significant difference between variance of both samples. [take F, 0.05(7,9) = 3.29] **05**

c. Fit a poisson distribution for following data and Test the Goodness of fit at 5% Level of significance. [given: $\chi_{2,0.05,4}$= 9.49] **05**

xi	0	1	2	3	4
fi	419	352	154	56	19

OR

6. a. Two types of batteries are tested for their length of life and following results were obtained. Battery A: n_1=10, $\bar{x}_1 = 500$ hours, $\sigma^2{}_1 = 100$, Battery B: n_2=10, $\bar{x}_2 = 500$ hours, $\sigma^2{}_2 = 121$. Compute t distribution and test whether there is a significant difference in two mean. [given $t_{0.05,18}$= 2.10] **04**

b. A correlation coefficient based on a sample of size 18 was computed to be 0.32. Can we conclude at 5% significance level that the corresponding population correlation coefficient is significant or we can say significantly different from zero? [Given $t_{0.025,16} = 2.120$]. **05**

c. A broker for a local investment firm has been studying the relationship between increase in price of gold (x) and her customer's requests to liquidate stocks (y). From data set based on 15 observations the sample slope was found to be 2.9. If the standard error of regression slope coefficient is 0.18, Is there any reason to believe that slope has changed from its past value of 3.2 at 5% level of significance. [given: $t_{0.025,13} = 2.160$]. **05**

7. a. What do you mean by transient state and steady-state? And Define Customer Behavior of queueing systems. **07**

b. Arrivals at a telephone booth are considered to be Poisson with an average time of 12 min. between one arrival and the next. The length of phone call is assumed to be distributed exponentially with mean 4 min. **07**

i. Find the average number of persons waiting in a system

ii. What is the probability that a person arriving at the booth will have to wait in the queue?

iii. What is the probability that it will take him more than 10 min. altogether to wait for the phone and complete his call?

iv. Estimate the fraction of the day when the phone will be in use.

v. The telephone department install a second booth, when convinced that an arrival has to wait on the average for atleast 3 min. for phone. By how much the flow of arrivals should increase in order to justify a second booth?

vi. What is the average length of the queue that forms time to time?

OR

8. a. What do the letters in the symbolic representation $(a/b/c : d/e)$ of a queuing model represent? **04**

b. A TV repairman finds that the time spent on his job has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which they came in and if the arrival of sets is approximately poisson with an average rate of 10 sets per 8 hr. in a day. **05**

i. What is the repairman's expected idle time each day?

ii. How many jobs are ahead of the average sets just brought in?

c. The arrival of customer at banking counter follows poisson distribution with a mean of 45 per hour. The service rate of the counter clerk also follows poisson distribution with a mean of 60 per hour. **05**

i. What is the probability of having zero customers in the system?

ii. What is the probability of having 5 customers in the system?

iii. Determine the steady statistics namely L_s, L_q, W_s, W_q